Siamese metric learning for generalised few-shot recognition of rare viruses in transmission electron microscopy images

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Abstract

Background and Objective: The public TEM virus dataset contains 22 virus classes, but its benchmark covers only the 14 most frequent; the eight rare viruses, with 5–25 training micrographs each, have not been studied as few-shot classes. We asked whether a Siamese metric-learning encoder can recognise these rare viruses without being trained on them.

Methods: A DenseNet201 encoder was trained with a supervised contrastive loss on the 14 base classes. Rare classes were added at test time as class prototypes (generalised few-shot learning). We compared it with the re-implemented benchmark classifier under identical, leakage-audited evaluation on the official split, selected every hyper-parameter on validation, and used micrograph-level cluster bootstrap and paired tests.

Results: On the 14 base classes the encoder reached $94.79 \pm 0.21\,\%$ accuracy (macro-sensitivity 0.946 ± 0.003, macro-specificity 0.9958 ± 0.0001) versus $93.91 \pm 0.54\,\%$ for the re-trained baseline; no seed pair differed significantly in accuracy. Over 22 classes, nearest-prototype classification gave macro-F1 0.786 ± 0.008. Confused rare classes were nearly linearly separable in the learned features, and logistic regression on the frozen features raised

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macro-F1 to 0.849 ± 0.009 (significant for every checkpoint). With one example per rare virus, Siamese features gave higher rare-class accuracy than baseline features (37.3 ± 2.3 % vs 30.8 ± 2.4 %; significant in all seed pairs); from three examples onwards, and with full rare-class data, they did not.

Conclusions: Siamese metric learning matches a strong classifier on this benchmark and helps in the one-shot regime. Most of the attainable gain on rare viruses comes from the decision rule rather than the representation.

Keywords: transmission electron microscopy, virus classification, few-shot learning, metric learning, rare classes

1. Introduction

Negative-stain transmission electron microscopy (TEM) can identify a virus without prior knowledge of what to look for, but interpreting the images requires scarce expertise. This has motivated automatic classification of virus particles in TEM images. Matuszewski and Sintorn released a public TEM virus dataset [1, 2]: 1,245 micrographs of 22 viruses, patches at 1 nm/pixel and an official, representative, micrograph-level train/validation/test split. Their benchmark uses the 14 most frequent classes; a fine-tuned DenseNet201 reached 93.1 % test accuracy. Eight classes were excluded because they had too few images or particles to train a classifier reliably.

Rare viruses are, however, the realistic diagnostic case: a newly emerging virus will always have few reference images. This is the setting of few-shot learning, and more precisely of generalised few-shot learning (GFSL), in which new classes must be recognised alongside the training classes [3]. Siamese metric learning fits it naturally: a shared-weight encoder maps

patches of the same virus close together, and a new virus is represented by the mean embedding (prototype) of its few examples, without retraining. Because few-shot methods are easily overrated when compared with weak baselines (Section 1.1), a credible evaluation must compare against a strong classification baseline under identical conditions.

To our knowledge no published work has applied metric-learning few-shot methods to this dataset or evaluated its rare classes as few-shot classes. Our contributions are: (i) a GFSL protocol for all 22 classes on the official split, in which the rare classes are never used for training and every choice is made on validation; (ii) a Siamese supervised-contrastive encoder compared with a faithful re-implementation of the benchmark classifier under identical evaluation; (iii) a diagnosis showing that rare-class errors stem from the nearest-prototype decision rule rather than the representation, and an evaluation of classifiers on frozen features that addresses them; and (iv) full reporting of negative results, with code, logs and per-patch predictions released.

1.1. Related work

Virus classification on the TEM virus dataset. The benchmark of Matuszewski and Sintorn [1] compared a custom CNN with eight ImageNet-pretrained architectures on the 14-class split and also showed that an unrepresentative split lowers performance substantially, which makes the official split the reference protocol. Later 14-class studies proposed ensembles of CNNs and vision transformers (96.7 % accuracy under an 80:20 protocol) [4], p4-equivariant group convolutions that improve data efficiency (84.49 % validation accuracy with the training set reduced to 93 particles

per class) [5], a compact dual-branch CNN fed with local-standard-deviation and discrete-cosine-transform images (97.44 % on a random 75/25 split of the augmented patch set) [6], and a DenseNet-based dual-track network with attention (99.73 %; class set and split not reported in the accessible text) [7]. Because these protocols differ, their accuracies are not directly comparable with each other or with the official benchmark. Two studies used all 22 classes, a dilated multilevel fused CNN (89.89 % accuracy) [8] and a CNN–transformer hybrid evaluated on a stratified split (88.31 %) [9]; both train a closed-set classifier on every class, and neither, as far as we could determine, reports rare-class results on the official split or treats the rare viruses as few-shot classes.

Siamese networks, metric learning and few-shot baselines. Siamese networks learn a shared-weight encoder and compare inputs by distance [10, 11]; prototypical networks extend the idea to N-way classification with class-mean prototypes [12]. Batch-based losses, namely the batch-hard triplet loss [13, 14] and the supervised contrastive (SupCon) loss [15], use every positive and negative pair in a class-balanced batch and train such encoders more efficiently than isolated pairs. Several studies have shown that a well-trained embedding with a simple classifier, such as nearest class mean after centring and normalisation [16] or logistic regression on frozen features [17], rivals more elaborate meta-learning [18], and that the advantage of meta-learning shrinks under strong domain shift [19]. Richer decision rules on frozen features include class-specific Mahalanobis distances with shrunk covariances [20] and distribution calibration, which borrows feature statistics from related base classes [21].

Generalised few-shot learning and medical imaging. In GFSL, new classes must be recognised among the training classes [3], and scores are typically biased towards the latter, which calibrated stacking counteracts [22]. In medical imaging, a systematic review of 80 few-shot studies found meta-learning to be the most common approach [23]. Few-shot recognition of viruses in TEM images, in which the rare classes are held out from training and evaluated on an official split, has, to our knowledge, not been studied.

2. Methods

2.1. Dataset and leakage control

We used the official patch folders of the dataset [2]: 256×256-pixel patches centred on annotated particles, after rescaling to 1 nm/pixel. The split assigns whole micrographs to train, validation or test (about 60/20/20 %) [1]. The 14 benchmark classes are the *base* classes; the eight excluded classes, which the dataset provides in the same split, are the *rare* classes (Table 1; examples in Fig. 1). The pre-augmented, class-balanced training set (736 patches per base class) was used only to re-implement the benchmark classifier, as in the original work.

Each patch was standardised to zero mean and unit variance [1] and its grey channel replicated to three channels; patches were not resized. Before training we audited the data at pixel level: none of the 16,441 training and augmented-training patches is identical to any rotation or reflection of a validation or test patch, and no test patch matches a validation patch.

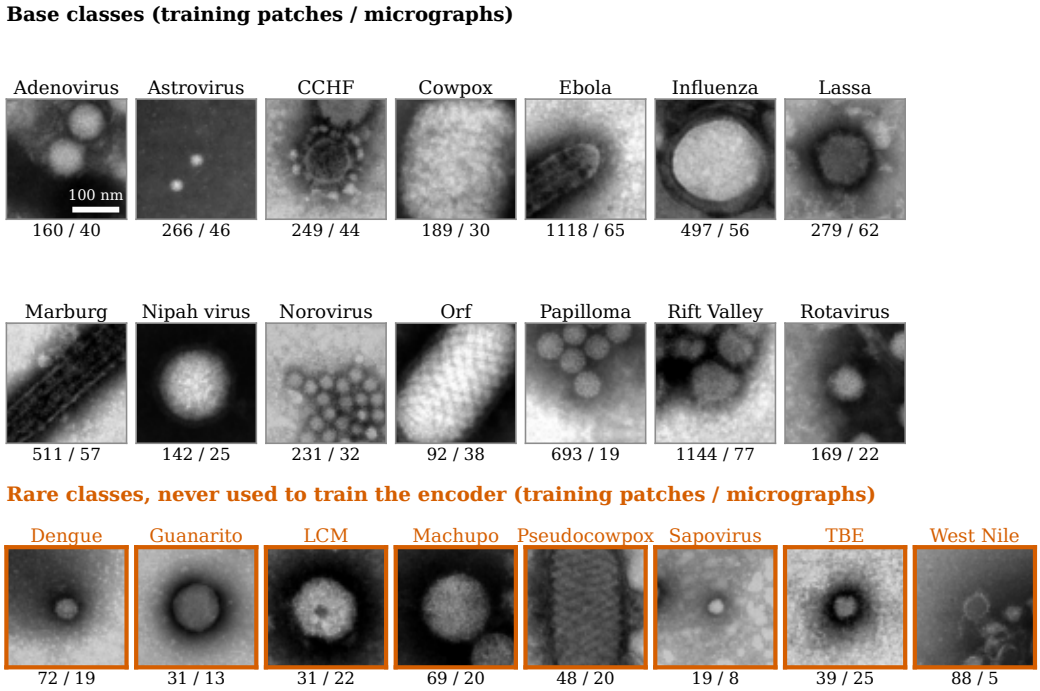


Figure 1: One example patch (256×256 pixels at 1 nm/pixel) of each virus from the official training split, with the number of training patches and micrographs. The eight rare classes (bottom, framed) are never used to train the encoder. Several are small, near-spherical particles of similar appearance, such as Dengue, Sapovirus and TBE. Patches from the TEM virus dataset [2] (CC BY 4.0).

2.2. Siamese metric-learning encoder

The encoder (Fig. 2a) is an ImageNet-pretrained DenseNet201 [24, 25]. Global average pooling gives a 1,920-dimensional *backbone feature*; a projection head (1,920→512→128, L2-normalised) produces the embedding used by the loss. All patches pass through the same weights, so similarities are computed in one shared space. The network was trained with the supervised contrastive loss [15] (temperature 0.1) on batches of 14 classes × 4 patches, drawing the patches of a class from different micrographs where possible.

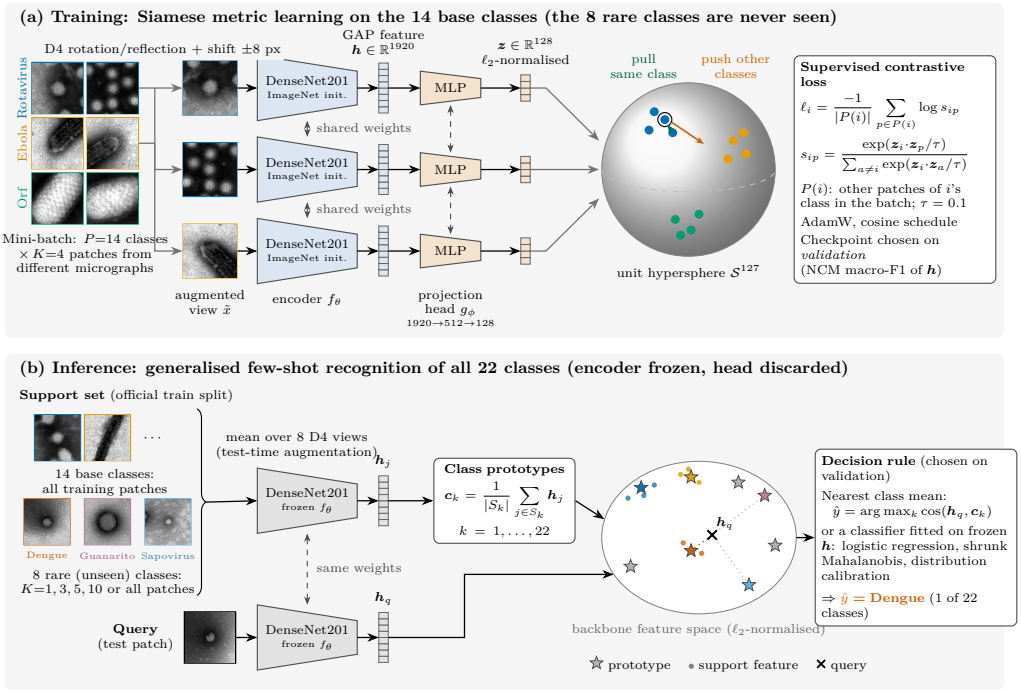


Figure 2: Overview of the method. (a) Training: patches of the 14 base classes pass through one DenseNet201 encoder with shared weights (drawn as parallel Siamese branches); a projection head maps the pooled feature $\boldsymbol{h}$ to a unit-norm embedding $\boldsymbol{z}$, and the supervised contrastive loss pulls together patches of the same virus and pushes apart patches of different viruses. (b) Inference: the head is discarded and the frozen encoder embeds the support patches of all 22 classes, including the eight rare classes never used for training, into class prototypes; a test patch is assigned by nearest class mean or by a classifier fitted on the frozen features, chosen on the validation split. Example patches are from the TEM virus dataset [2] (CC BY 4.0).

Augmentation used the eight exact rotations and reflections of the square and translations of up to ±8 pixels, which preserve physical scale. We used AdamW [26] (learning rates 5×10^{-5} for the backbone and 5×10^{-4} for the head, weight decay 10^{-4}, cosine schedule, at most 40 epochs). The check-

point with the highest validation macro-F1 of nearest-prototype classification over the base classes was kept, and training stopped after 12 epochs without improvement (training curves in Fig. B.5). The rare classes were never seen during training or model selection.

Pilot experiments in which four base classes were withheld from training showed that the backbone feature transfers to unseen classes whereas the 128-dimensional head does not (Appendix A). We therefore fixed the backbone feature as the representation for all experiments before evaluating the main models.

2.3. Baselines

B1 re-implements the best benchmark model [1] from its published configuration and code: DenseNet201, two fully connected layers of 1,024 units with dropout 0.3, and a 14-way softmax, trained with cross-entropy on the augmented training set (Adam, learning rate 10^{-5}, batch size 12, 50 epochs, best validation accuracy). B1 and the Siamese encoder were each trained with 3 seeds. B1's pooled DenseNet201 features also serve as an alternative backbone in every few-shot experiment, which isolates the effect of the training objective. **B2**, a DenseNet201 with a linear head trained with class-balanced sampling on all 22 classes, sees the rare classes during training and is reported only as a fully supervised reference. Two loss ablations were trained once: batch-hard triplet loss [14] and SupCon plus an auxiliary cosine-classifier term [18].

2.4. Evaluation protocols

All results are on the official test split, with each patch represented by the mean over its eight rotations and reflections, computed with full-precision (fp32), deterministic inference. *Closed set (14 classes):* nearest class mean with cosine similarity, prototypes from all training patches; B1 uses its softmax. *GFSL (22 classes; Fig. 2b):* prototypes for all classes from the training split; the 2,057 test patches are classified among all 22 classes. We report accuracy, macro-F1, base-class accuracy and rare-class mean per-class accuracy; the last is preferred to pooled accuracy because rare test sets range from 7 to 44 patches. *K-shot GFSL:* as GFSL, but each rare class receives only $K \in \{1, 3, 5, 10\}$ training patches from different micrographs (100 random draws); base classes keep all training patches. Episode and unknown-virus detection results are given in Appendix B.

2.5. Classifiers on frozen features

To test whether rare-class errors stem from the representation or from the decision rule, we fitted four classifiers on the frozen backbone features of every checkpoint, without modifying the encoder: nearest class mean (NCM); L2-regularised logistic regression (LR) with class-balanced weights [17]; a Mahalanobis classifier with class covariances shrunk towards the pooled covariance in a 64–256-dimensional principal-component space, following Simple CNAPS [20]; and distribution calibration (DC), which samples synthetic features for each rare class from the statistics of its nearest base classes [21]. Each classifier was fitted on the training split and its hyper-parameters chosen by 22-class validation macro-F1; the chosen setting was applied once to the test split. Before seeing any test result we fixed the rule that the classifier

with the highest validation macro-F1 is the one reported as the improvement; all four are reported. In a separability diagnostic, a binary LR was trained on the training patches of each of eight confusable pairs and evaluated on their validation patches.

2.6. Statistics

Confidence intervals (95 %) come from a stratified micrograph-level cluster bootstrap that resamples micrographs within classes [27]. Paired comparisons on identical test patches use a paired cluster bootstrap of the difference in accuracy and macro-F1, together with an exact McNemar test [28]. McNemar treats patches as independent and is anti-conservative when patches share micrographs, so conclusions rest on the cluster bootstrap. Siamese–B1 comparisons are made for all 3×3 seed pairs, and K-shot comparisons are paired over identical support draws. Values over seeds are mean $\pm$ standard deviation.

2.7. Implementation

Experiments used PyTorch 2.11 and scikit-learn on one NVIDIA RTX 3060 (12 GB). The analysis code was written with the assistance of an AI tool (Claude, Anthropic) and was checked with 23 automated tests, including comparisons of the loss functions and evaluation metrics against independent reference implementations. Training took 19–28 min per Siamese seed and about 134 min for B1.

3. Results

3.1. 14-class benchmark

The Siamese encoder matched B1 (Table 2). Over the 9 Siamese–B1 seed pairs, the paired accuracy difference ranged from +0.05 to +1.47 points; it favoured the Siamese encoder in 9 pairs, but its cluster CI excluded zero in 0 of 9 (macro-F1: +0.000 to +0.021, CI excluding zero in 1 of 9). For seed 0 versus seed 0, the accuracy difference was +0.47 points (95 % CI [-1.21, +2.04]; McNemar 48 vs 39 discordant patches, $p = 0.39$). Every seed of both models exceeded the published 93.1 %; because test-time augmentation and implementation details differ, B1 rather than the published figure is the fair comparator. The 128-dimensional head scored slightly higher on the base classes (95.72 ± 0.35 %), but it was not the pre-specified representation, its CI against B1 (seed 0) includes zero ([-0.38, +3.06]), and it fails on unseen classes (Section 3.2).

3.2. 22-class generalised few-shot recognition

With nearest-prototype classification over all 22 classes, Siamese and B1 features reached the same macro-F1 (0.786 ± 0.008 vs 0.785 ± 0.008; Table 3). Siamese features were more accurate overall (89.56 ± 0.36 % vs 87.57 ± 0.38 %), mainly through the base classes (91.8 ± 0.3 % vs 89.4 ± 0.6 %), whereas B1 features recognised the rare classes slightly better (rare mean-class accuracy 77.4 ± 1.7 % vs 74.5 ± 2.7 %). Over the 9 seed pairs the paired accuracy difference ranged from +1.36 to +2.72 points, with a cluster CI excluding zero in 2 of 9; for seed 0 the McNemar test gave $p = 0.00064$ although the cluster CI [-0.54, +4.72] includes zero, illustrating its anti-conservatism.

The 128-dimensional head collapsed on the rare classes (rare mean-class accuracy 33–43 %). A large gap remained to B2, which was trained on the rare classes. Errors concentrated in a few rare viruses: 32–35 of the 44 Dengue test patches were assigned to Sapovirus in every seed.

3.3. Rare classes are separable

Every confusable pair was close to linearly separable in the learned features (validation AUROC ≥ 0.94; Table 4), yet the two-class nearest-mean rule was far below the linear boundary for several pairs. For Sapovirus versus Astrovirus it was near chance (0.46–0.53) while LR reached 0.86–0.91. The information needed to recognise the rare viruses is therefore present in the representation but not exploited by isotropic distance to a class mean. The rare-class validation sets are small (10–16 patches), so these values are indicative.

3.4. Classifiers on frozen features

The pre-declared rule selected LR for 6 of the 7 checkpoints, including all three Siamese seeds (exception: shrunk Mahalanobis for B1 seed 1; Table 5). For the Siamese encoder, LR raised 22-class macro-F1 from 0.786 ± 0.008 to 0.849 ± 0.009 and accuracy from 89.56 ± 0.36 % to 92.72 ± 0.27 %, while base-class accuracy also rose (91.8 ± 0.3 to 94.6 ± 0.1 %). The paired macro-F1 gain of LR over NCM ranged from +0.049 to +0.091 and its cluster CI excluded zero for 7 of the 7 checkpoints. Dengue F1 rose from 0.12 to 0.34, Sapovirus from 0.20 to 0.35 and Guanarito from 0.43 to 0.77 (Appendix Table B.7). The gain came mainly from precision (Fig. 3): averaged over the eight rare classes, precision rose from 0.49 to 0.69 and recall from 0.74

to 0.80. With NCM, rare prototypes attracted many patches of other classes (precision 0.11 for Sapovirus and 0.28 for Guanarito). Dengue remained the hardest class: 77 % of its test patches were assigned to Sapovirus with NCM and still 52 % with LR (recall 10 % and 26 %). The Mahalanobis classifier reached a similar macro-F1 (0.861 ± 0.008), but its validation optimum was nearly flat across settings (smallest gap between the best two settings 0.0006), so its test result depends on near-ties; we report it as secondary.

With full rare-class training data the Siamese objective gave no advantage: LR on B1 features reached macro-F1 0.855 ± 0.013 versus 0.849 ± 0.009 for Siamese features. Across the 9 seed pairs the paired macro-F1 difference (Siamese minus B1) ranged from -0.026 to +0.015, with a CI excluding zero in 1 of the 9 pairs (1 in favour of B1).

Classifiers that estimate more parameters needed more data. With one example per rare class, LR fell to 18.0 ± 0.8 % rare mean-class accuracy against 36.9 ± 2.6 % for NCM, whereas DC improved on NCM (43.5 ± 2.4 %; paired gain +4.8 to +17.2 points, significant for 6 of 6 Siamese and B1 checkpoints). With DC, Siamese and B1 features performed similarly at one shot (43.5 ± 2.4 % vs 45.8 ± 3.1 %).

3.5. Few-shot regime

With one training patch per rare virus, Siamese features gave higher rare-class accuracy than B1 features (37.3 ± 2.3 % vs 30.8 ± 2.4 %; Fig. 4); paired over identical support draws, the difference ranged from +2.9 to +11.9 points and was significant for 9 of 9 Siamese–B1 pairs. From three examples onwards B1 features were equal or better (numerical values in Appendix Table B.9).

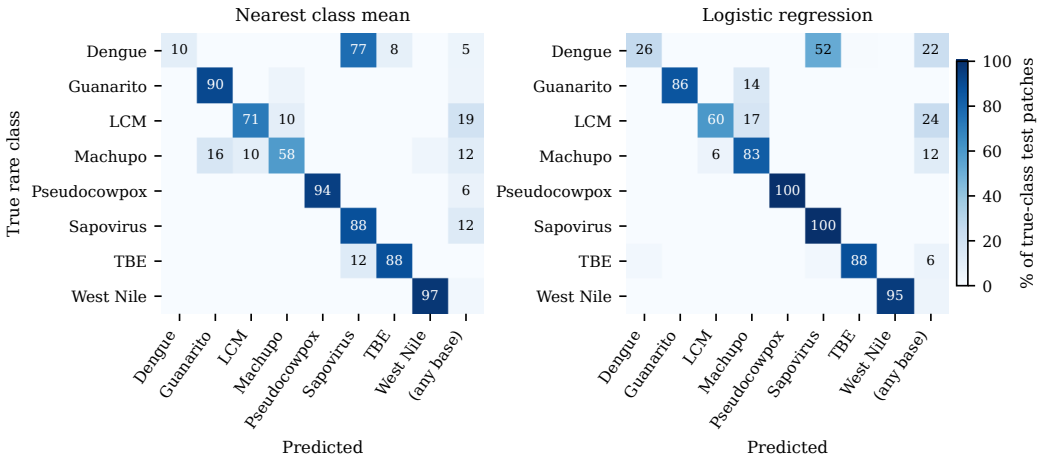


Figure 3: Where the rare-class test patches go, for Siamese features with nearest class mean (left) and with logistic regression (right). Rows: true rare class; columns: predicted rare class, or any of the 14 base classes pooled. Counts are summed over the three seeds and normalised per row (%); values below 5 % are not printed.

4. Discussion

Under an identical, leakage-audited protocol, a Siamese supervised-contrastive encoder with a nearest-prototype classifier matched a re-implementation of the strongest published benchmark model on the 14 base classes, and its features allow new virus classes to be added without retraining. In the most data-scarce setting, a single reference patch per new virus classified by nearest prototype, it recognised rare viruses better than features learned with cross-entropy, consistently across all seed pairs. This is the regime that motivates few-shot learning for diagnostic TEM: the first image of an emerging virus can be added as a prototype immediately.

The advantage did not persist with three or more examples, in rare-class-only episodes, when distribution calibration was used, or once a linear

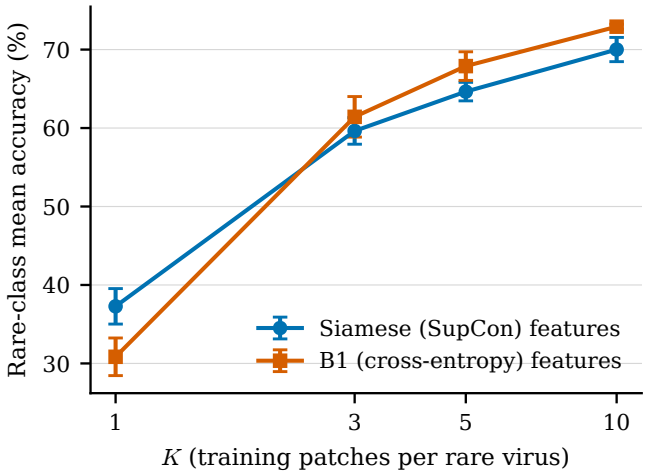


Figure 4: K-shot GFSL with nearest class mean: rare-class mean accuracy when each rare class has only K training patches (from different micrographs), while the 14 base classes keep all training patches. Mean ± SD over three seeds per encoder; each seed averages 100 random support draws.

classifier was fitted on all available rare-class data; with nearest-prototype classification over all 22 classes the two feature sets also reached the same macro-F1. This agrees with the finding that the quality of the embedding, rather than the metric-learning objective, dominates few-shot performance [18, 17], and it shows that claims of superiority for metric-learning approaches on this dataset should be tested against an identically evaluated classification baseline.

The most actionable result concerns the rare classes that previous 22-class work did not analyse. Dengue, Sapovirus, Guanarito and LCM were poorly recognised by prototype classification although they are nearly linearly separable from their confusers in the same feature space. Distance to a class mean ignores differences in class spread, which is plausible for rare

classes whose few micrographs differ in staining and particle density. A validation-selected logistic regression recovered much of the gap to the fully supervised reference B2 (macro-F1 0.849 ± 0.009 vs 0.897) without retraining the encoder. It needs enough examples, however: with one to five examples per class, nearest-mean or distribution calibration should be preferred.

Published 22-class results train on all classes and use different or unreported splits [8, 9], so they are not directly comparable with ours; B2, trained on all classes with our pipeline, reached 93.10 % accuracy on the official split.

Limitations. The rare classes have 2–8 test micrographs each, so per-class F1 for Sapovirus (8 patches) and Guanarito (7 patches) is highly uncertain. The ablations were trained once. Hyper-parameters of the frozen-feature classifiers were chosen on small rare-class validation sets, which adds selection variance, as the Mahalanobis results show. We studied one backbone and one dataset, and unknown-virus detection (Appendix B) is not yet adequate for deployment.

5. Conclusions

To our knowledge, this is the first generalised few-shot evaluation of all 22 classes of the public TEM virus dataset on its official split, with the rare classes never used for training. A Siamese metric-learning encoder matches a strong re-implemented classifier on the benchmark and, with nearest-prototype classification, recognises rare viruses better when only one reference patch is available, but not with more examples. Most of the attainable improvement on rare viruses comes from replacing nearest-prototype classification with a validation-selected linear classifier on frozen features, for

either training objective. The protocol, code and per-patch predictions are released for future comparisons.

Declaration of competing interests

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRediT authorship contribution statement

S. M. Mahbub Uddin Ahmed: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Data and code availability

The TEM virus dataset is publicly available [2]. Code, configurations, training logs, evaluation outputs and per-patch predictions are available at `https://github.com/hrid0yyy/tem-virus-siamese-fewshot` [TODO before submission: repository is still private – make it public or change this URL]; trained checkpoints are available as release assets.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) in order to assist with writing the analysis code and with drafting and editing the manuscript text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Appendix A. Pilot experiments

Four base classes were withheld from encoder training (F2: Astrovirus, Influenza, Papilloma, Rotavirus; F3: Marburg, Orf, Adenovirus, CCHF) and used as unseen classes, with micrograph-disjoint episodes (600 episodes, 15 queries per class). Without training, ImageNet features reached only 66.0 % nearest-mean accuracy on the 14 classes (88.1 % with a linear probe). After training, the backbone feature transferred to the unseen classes, whereas the 128-dimensional head did not; in the F3 GFSL test head accuracy on the unseen classes was 26.9 % (Table A.6).

Appendix B. Additional results

Training curves of the three Siamese and three B1 runs are shown in Fig. B.5. The Siamese runs stopped early when validation macro-F1 had not improved for 12 epochs; B1 was trained for the fixed 50 epochs of the original configuration, and its validation accuracy fluctuated from epoch to epoch, so the selected checkpoint (best validation accuracy) varies between seeds.

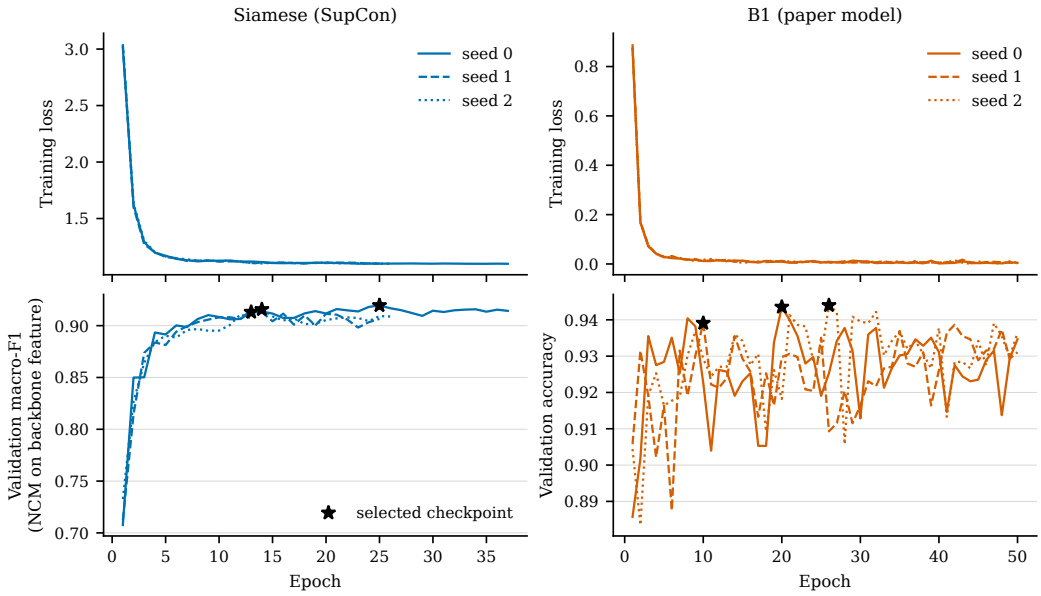


Figure B.5: Training loss (top) and the validation metric used for checkpoint selection (bottom) per epoch for the three seeds of each encoder. Stars mark the selected checkpoints.

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Table 1: Patches (micrographs) per class in the official split.

Class	Role	Train	Validation	Test
Adenovirus	base	160 (40)	44 (13)	86 (14)
Astrovirus	base	266 (46)	86 (15)	66 (15)
CCHF	base	249 (44)	100 (15)	86 (15)
Cowpox	base	189 (30)	72 (10)	59 (10)
Ebola	base	1,118 (65)	469 (22)	363 (22)
Influenza	base	497 (56)	281 (19)	170 (19)
Lassa	base	279 (62)	114 (21)	128 (21)
Marburg	base	511 (57)	174 (19)	173 (19)
Nipah virus	base	142 (25)	40 (8)	35 (8)
Norovirus	base	231 (32)	104 (11)	84 (11)
Orf	base	92 (38)	76 (13)	31 (13)
Papilloma	base	693 (19)	227 (6)	187 (6)
Rift Valley	base	1,144 (77)	414 (26)	392 (26)
Rotavirus	base	169 (22)	48 (7)	40 (7)
Dengue	rare	72 (19)	15 (6)	44 (7)
Guanarito	rare	31 (13)	10 (4)	7 (4)
LCM	rare	31 (22)	10 (7)	14 (8)
Machupo	rare	69 (20)	16 (6)	23 (7)
Pseudocowpox	rare	48 (20)	21 (7)	16 (7)
Sapovirus	rare	19 (8)	11 (3)	8 (3)
TBE	rare	39 (25)	12 (8)	11 (8)
West Nile	rare	88 (5)	66 (2)	34 (2)
Total		6,137 (745)	2,410 (248)	2,057 (252)

Table 2: 14-class official test split (1,900 patches, 206 micrographs). Macro-averaged precision, sensitivity (recall) and specificity; mean ± SD over seeds. [a]Without test-time augmentation.

Model	Acc. (%)	Macro-F1	Prec.	Sens.	Spec.	Micrograph acc. (%)
DenseNet201, published [1][a]	93.10	0.921	0.926	0.921	–	–
B1: same model re-trained here (3 seeds)	93.91 ± 0.54	0.930 ± 0.007	0.927 ± 0.006	0.939 ± 0.013	0.9952 ± 0.0005	95.95 ± 0.74
Siamese SupCon, backbone feature (3 seeds)	94.79 ± 0.21	0.943 ± 0.004	0.944 ± 0.005	0.946 ± 0.003	0.9958 ± 0.0001	96.60 ± 0.49
Ablation: batch-hard triplet (1 seed)	93.58	0.929	0.933	0.928	0.9948	96.12
Ablation: SupCon + cosine CE (1 seed)	94.37	0.938	0.947	0.934	0.9954	96.60
Siamese 128-d head (not pre-specified)	95.72 ± 0.35	0.952 ± 0.003	0.955 ± 0.005	0.951 ± 0.003	0.9965 ± 0.0003	96.93 ± 0.28

Table 3: 22-class GFSL on the official test split (2,057 patches, 252 micrographs) with nearest-class-mean classification; rare classes enter only as prototypes.

Features	Acc. (%)	Macro-F1	Base acc. (%)	Rare mean-class acc. (%)
Siamese backbone feature (3 seeds)	89.56 ± 0.36	0.786 ± 0.008	91.8 ± 0.3	74.5 ± 2.7
B1 backbone feature (3 seeds)	87.57 ± 0.38	0.785 ± 0.008	89.4 ± 0.6	77.4 ± 1.7
Siamese 128-d head	89.35 ± 0.18	0.693 ± 0.013	94.1 ± 0.1	39.1 ± 5.1
Ablation: batch-hard triplet	88.67	0.779	90.7	76.4
Ablation: SupCon + cosine CE	89.35	0.794	91.2	79.4
Reference B2: trained on all 22 classes	*93.10*	*0.897*	*95.1*	*81.8*

Table 4: Pairwise separability on the validation split (binary LR trained on training patches). Values are means over the three seeds of each encoder.

	Siamese features			B1 features	
Pair	AUROC	LR acc.	NCM acc.	AUROC	Val. patches
Dengue vs Sapovirus	1.000	1.00	0.91	1.000	15/11
Dengue vs TBE	0.974	0.95	0.80	0.981	15/12
Dengue vs Astrovirus	1.000	0.95	0.87	1.000	15/86
Sapovirus vs Norovirus	0.993	0.99	0.99	0.993	11/104
Sapovirus vs Astrovirus	0.957	0.87	0.49	0.920	11/86
Guanarito vs Lassa	0.986	0.96	0.85	0.989	10/114
Guanarito vs Machupo	0.958	0.86	0.67	0.948	10/16
LCM vs Machupo	0.979	0.91	0.87	0.990	10/16

Table 5: 22-class GFSL on the official test split with classifiers on frozen backbone features; hyper-parameters selected on validation. $K = 1$, $K = 5$: rare mean-class accuracy with only K training patches per rare class (30 draws). Superscripts: number of seeds for which the classifier had the highest validation macro-F1.

Features	Classifier	Val. F1	Acc. (%)	Macro-F1	Rare (%)	K=1 (%)	K=5 (%)
Siamese	Nearest class mean	0.758 ± 0.019	89.56 ± 0.36	0.786 ± 0.008	74.5 ± 2.7	36.9 ± 2.6	65.6 ± 1.5
	Logistic regression$^{3/3}$	0.851 ± 0.011	92.72 ± 0.27	0.849 ± 0.009	79.6 ± 1.0	18.0 ± 0.8	58.9 ± 0.7
	Shrunk Mahalanobis	0.838 ± 0.008	92.56 ± 0.69	0.861 ± 0.008	75.0 ± 4.8	25.3 ± 4.8	47.5 ± 8.7
	Distribution calibration	0.754 ± 0.012	89.95 ± 0.65	0.776 ± 0.008	73.8 ± 0.6	43.5 ± 2.4	64.4 ± 2.2
B1	Nearest class mean	0.766 ± 0.020	87.57 ± 0.38	0.785 ± 0.008	77.4 ± 1.7	29.7 ± 2.0	67.6 ± 1.8
	Logistic regression$^{2/3}$	0.872 ± 0.012	92.55 ± 0.67	0.855 ± 0.013	79.5 ± 1.8	14.6 ± 1.4	57.0 ± 2.0
	Shrunk Mahalanobis$^{1/3}$	0.859 ± 0.017	91.33 ± 0.29	0.846 ± 0.001	76.1 ± 1.8	15.5 ± 1.2	49.6 ± 2.1
	Distribution calibration	0.754 ± 0.017	88.59 ± 0.71	0.766 ± 0.010	76.9 ± 6.1	45.8 ± 3.1	69.1 ± 5.8

Table A.6: Pilot experiments (accuracy in %, or harmonic mean of base and unseen accuracy).

Experiment	Frozen ImageNet	Trained, backbone feature	Trained, 128-d head
F2: 4-way 5-shot on held-out classes	86.9	96.9 (+10.0 ± 0.4)	84.8
F3: 4-way 5-shot on held-out classes	91.2	96.0 (+4.8 ± 0.3)	93.1
F3: GFSL harmonic mean (10 base + 4 held-out)	0.676	0.843	0.419

Table B.7: Test F1 of the weakest rare classes. Test patches: Dengue 44, Sapovirus 8, Guanarito 7, LCM 14, Machupo 23.

Features, classifier	Dengue	Sapovirus	Guanarito	LCM	Machupo
Siamese, Nearest class mean	0.12	0.20	0.43	0.49	0.62
Siamese, Logistic regression	0.34	0.35	0.77	0.64	0.80
Siamese, Shrunk Mahalanobis	0.44	0.26	0.82	0.78	0.81
Siamese, Distribution calibration	0.15	0.22	0.47	0.48	0.62
B1, Nearest class mean	0.12	0.20	0.51	0.51	0.68
B1, Logistic regression	0.30	0.35	0.86	0.68	0.76
B1, Shrunk Mahalanobis	0.38	0.27	0.77	0.65	0.75
B1, Distribution calibration	0.16	0.20	0.47	0.48	0.61

Table B.8: 8-way episodes over the rare classes (micrograph-disjoint support and query, 600 episodes) and unknown-virus detection (rare-class test patches as unknown; score = maximum cosine similarity to the 14 base prototypes).

Features	1-shot (%)	5-shot (%)	AUROC	FPR at 95 % TPR (%)
Siamese (3 seeds)	65.0 ± 1.3	78.2 ± 1.0	0.883 ± 0.003	58.2 ± 3.1
B1 (3 seeds)	65.7 ± 0.8	80.3 ± 1.0	0.881 ± 0.015	60.7 ± 5.8
SupCon + cosine CE	65.6	79.0	0.899	52.9
Batch-hard triplet	62.9	77.1	0.868	70.7

Table B.9: Numerical values of Fig. 4: rare-class mean accuracy (%) with K training patches per rare class, nearest class mean, mean ± SD over three seeds (100 draws per seed).

K	Siamese features	B1 features
1	37.3 ± 2.3	30.8 ± 2.4
3	59.6 ± 1.7	61.4 ± 2.6
5	64.6 ± 1.2	67.9 ± 1.8
10	70.0 ± 1.5	72.9 ± 0.3